```
Nigeria, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277             -36452.2            -36449.2                 0
     2 20263017             5000           0.276             -36460.4            -36444.0                 0
     3 20263018             5000           0.274             -36453.1            -36448.7                 0
     4 20263019             5000           0.274             -36463.3            -36447.6                 0
                                    file
 mcmc_nigeria_step2_chain1_2026-10-19.Rdata
 mcmc_nigeria_step2_chain2_2026-10-20.Rdata
 mcmc_nigeria_step2_chain3_2026-10-20.Rdata
 mcmc_nigeria_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
   age.30.to.39.transmission.multiplier.0    age.20.to.29.transmission.multiplier.0           hiv.specific.mortality.rates.1
                                   2.772                                   2.337                                    2.203
   age.30.to.39.transmission.multiplier.3    age.15.to.19.transmission.multiplier.3    age.15.to.19.transmission.multiplier.0
                                   1.968                                   1.955                                    1.872
          hiv.specific.mortality.rates.0    age.40.to.49.transmission.multiplier.0                                 trate.3
                                   1.807                                   1.805                                    1.787
                                 trate.0   age.25.to.49.hiv.mortality.multiplier.1                       age.assortativity
                                   1.743                                   1.741                                    1.615
age.50.and.over.transmission.multiplier.3   age.25.to.49.hiv.mortality.multiplier.0    age.20.to.29.transmission.multiplier.3
                                   1.604                                   1.603                                    1.520
                female.to.male.multiplier                                 trate.4                                  trate.2
                                   1.484                                   1.443                                    1.437
         male.hiv.mortality.multiplier.0    age.40.to.49.transmission.multiplier.1                male.awareness.multiplier
                                   1.411                                   1.341                                    1.327
   age.40.to.49.transmission.multiplier.2                                 trate.1    age.30.to.39.transmission.multiplier.2
                                   1.310                                   1.309                                    1.286
over.80.mortality.slope.multiplier.female
                                   1.277

Merged simset:
 location simulations total.weight                                                file
  Nigeria        1000         1000 simset_nigeria_step2_chains1234_2026-10-20.Rdata
```

Nigeria step 2: likelihood

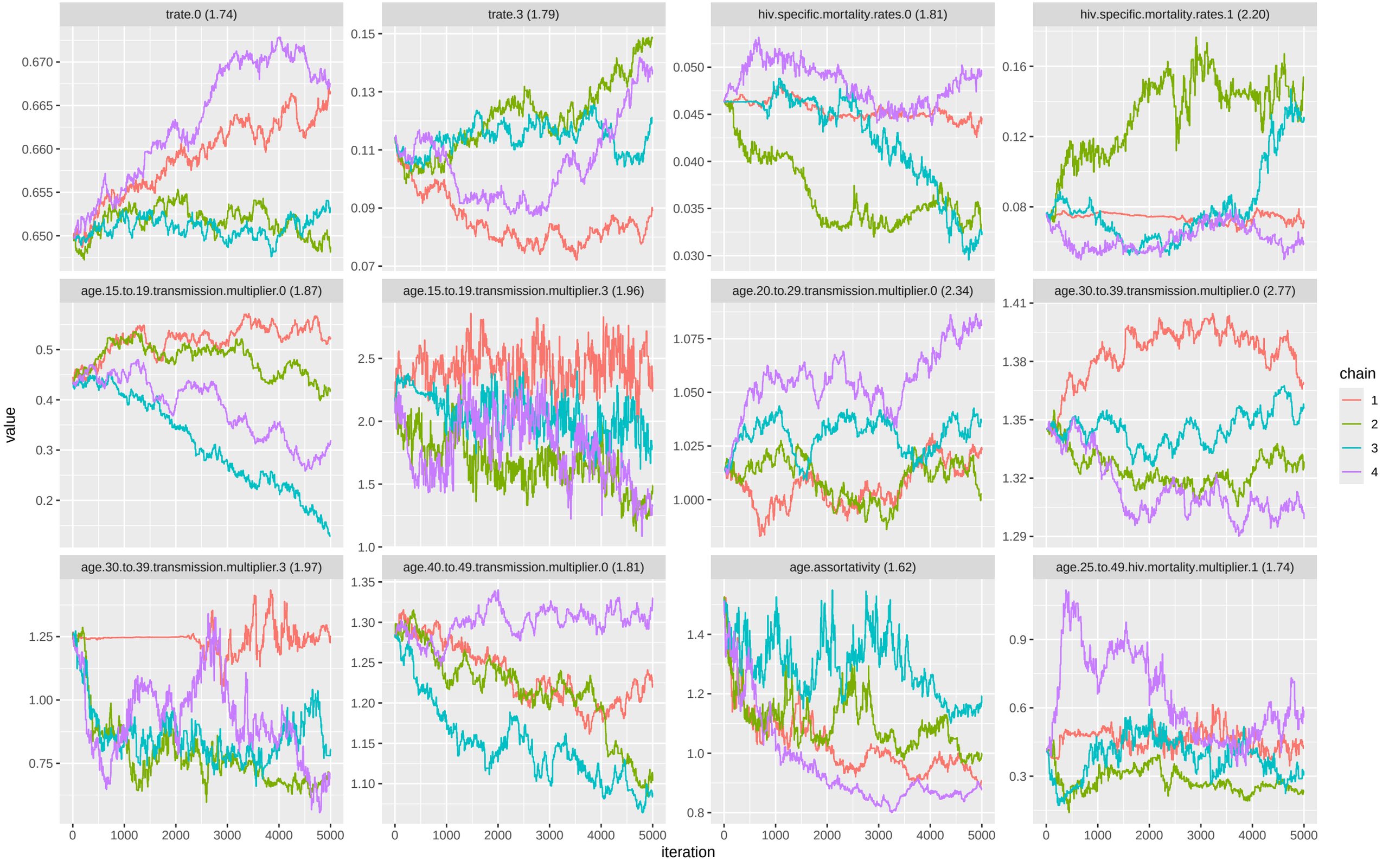
0
−10000
−20000
−30000
lue

0
25000
50000
75000
100000
iteration

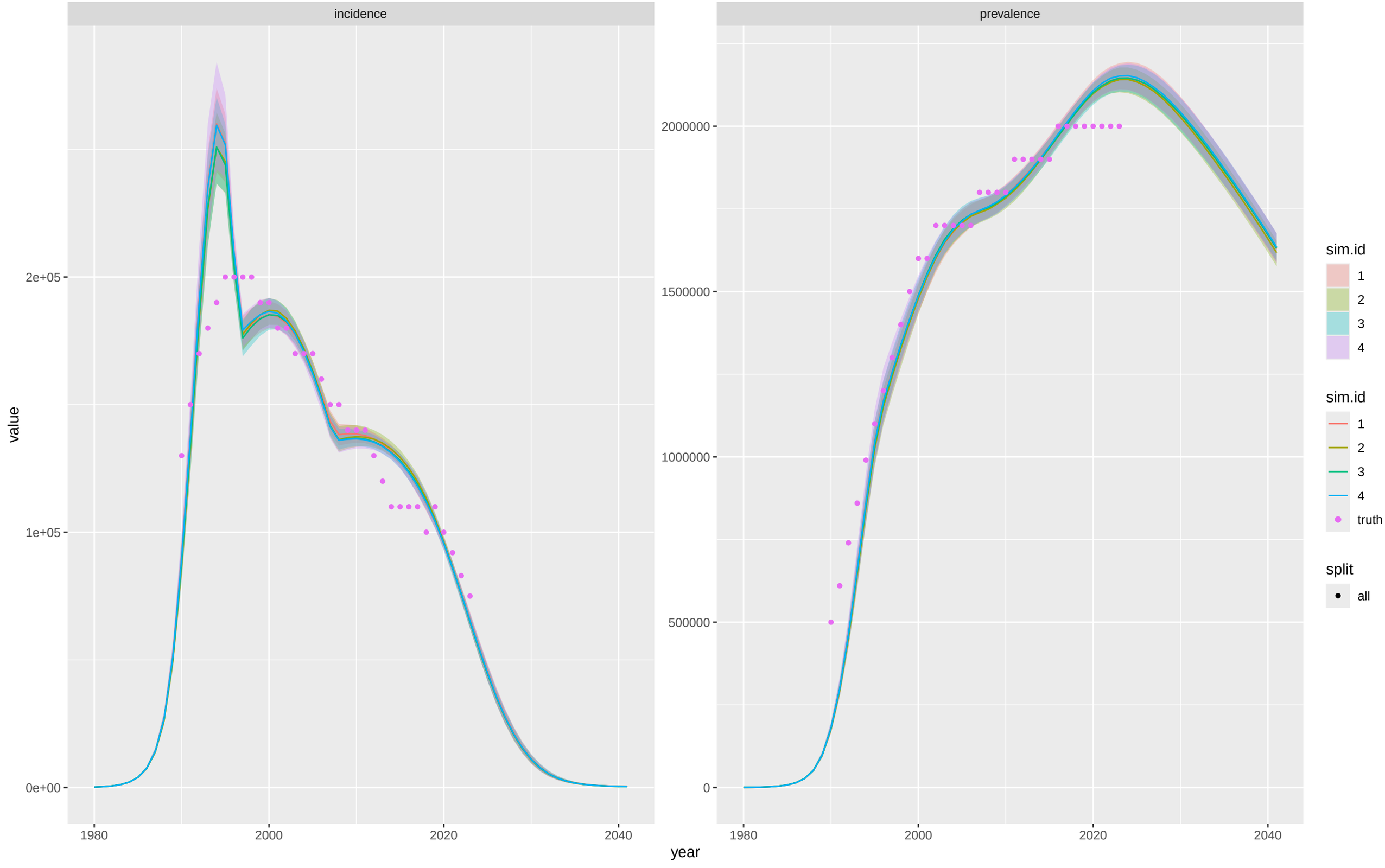
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

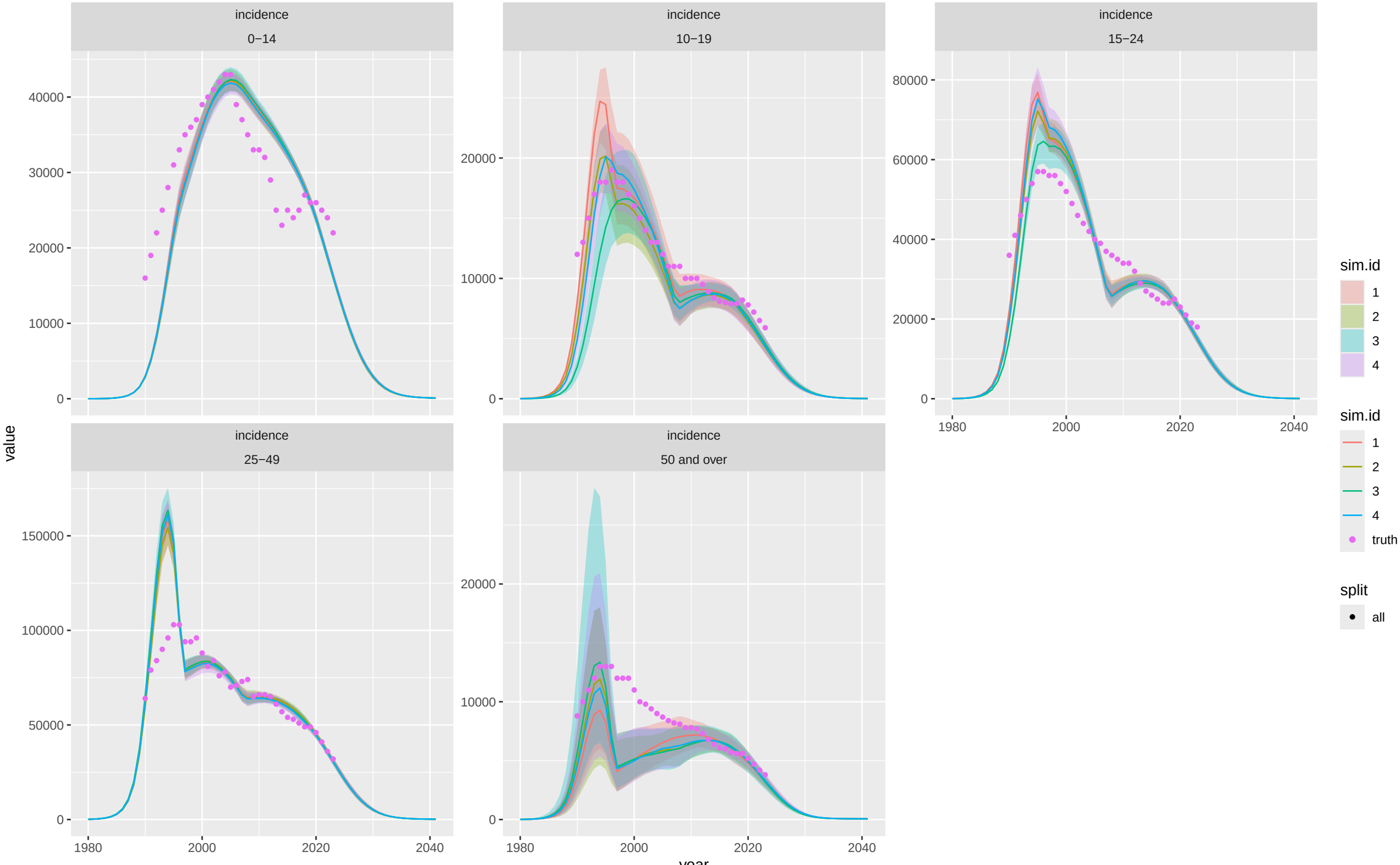
Nigeria step 2: trace



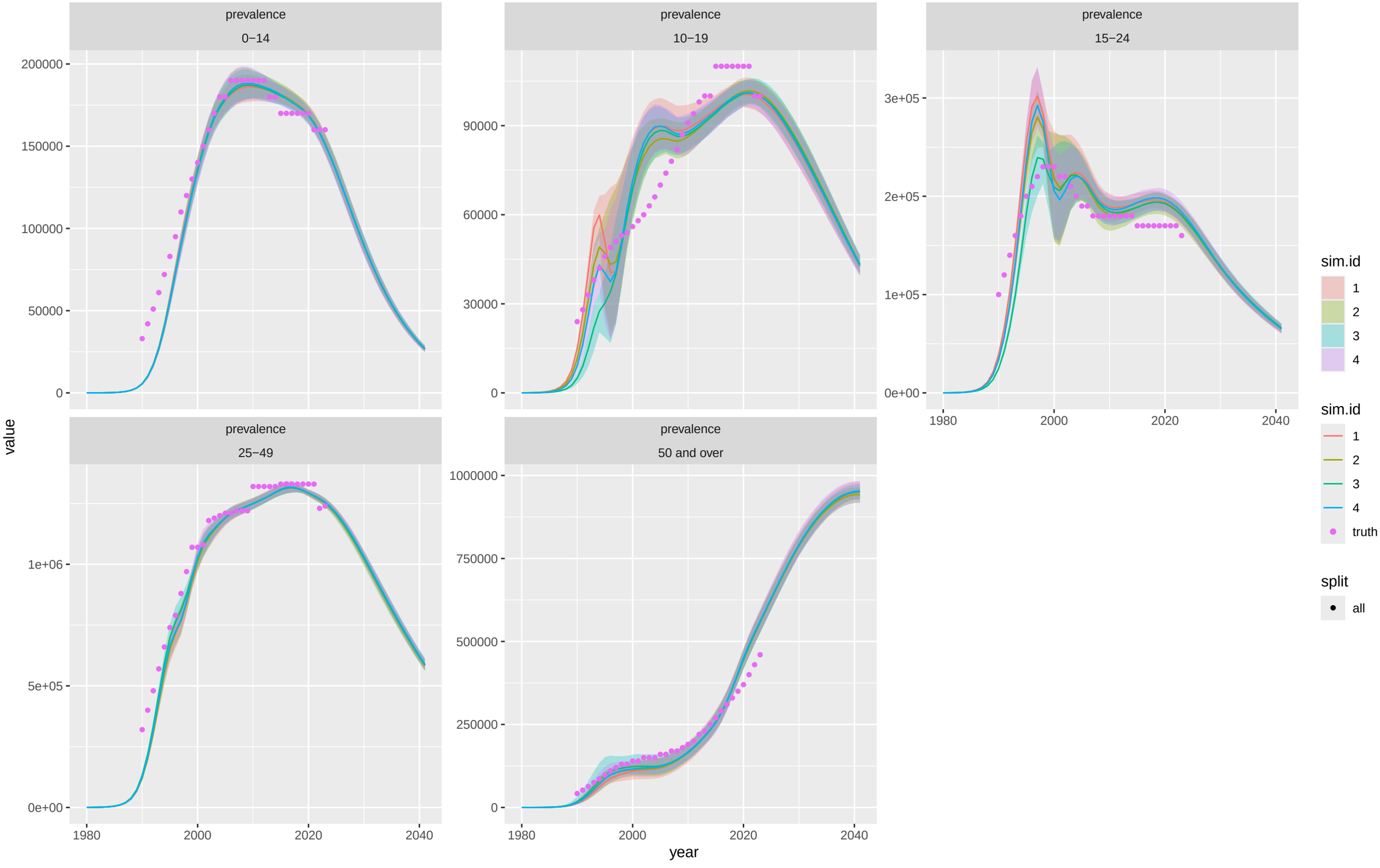
Nigeria step 2: fit (one color per chain)



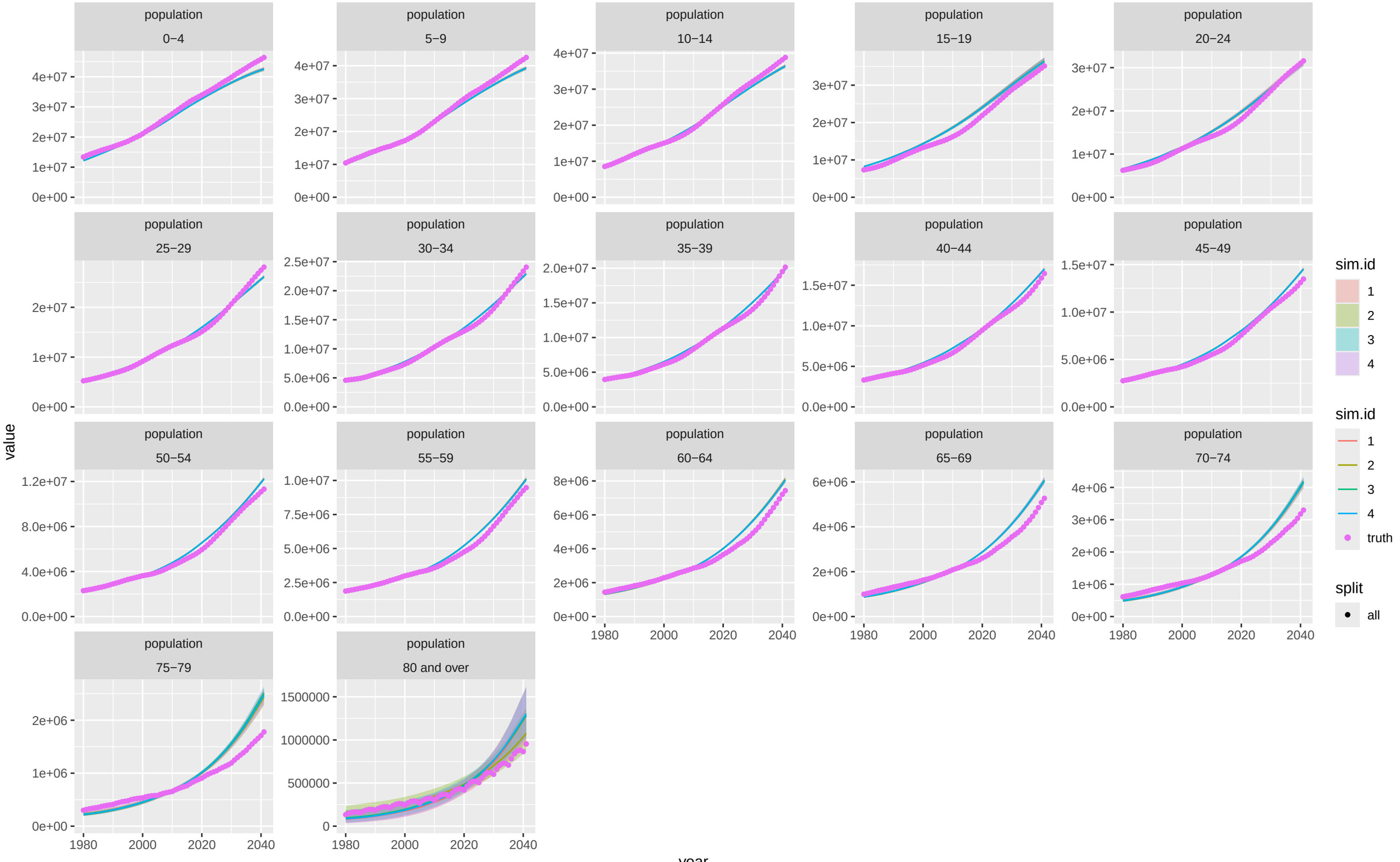
Nigeria step 2: incidence.by.age (one color per chain)



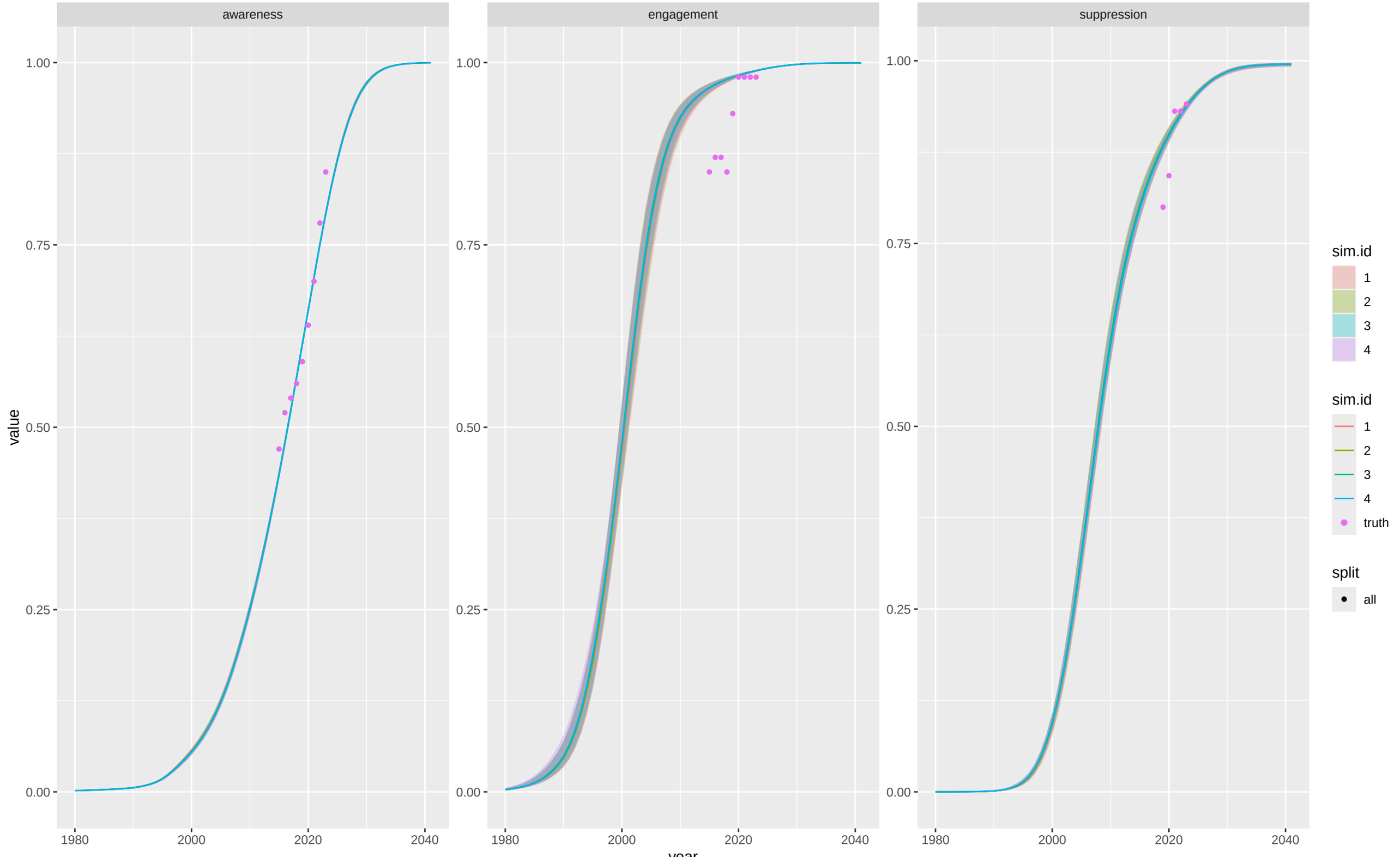
Nigeria step 2: prevalence.by.age (one color per chain)



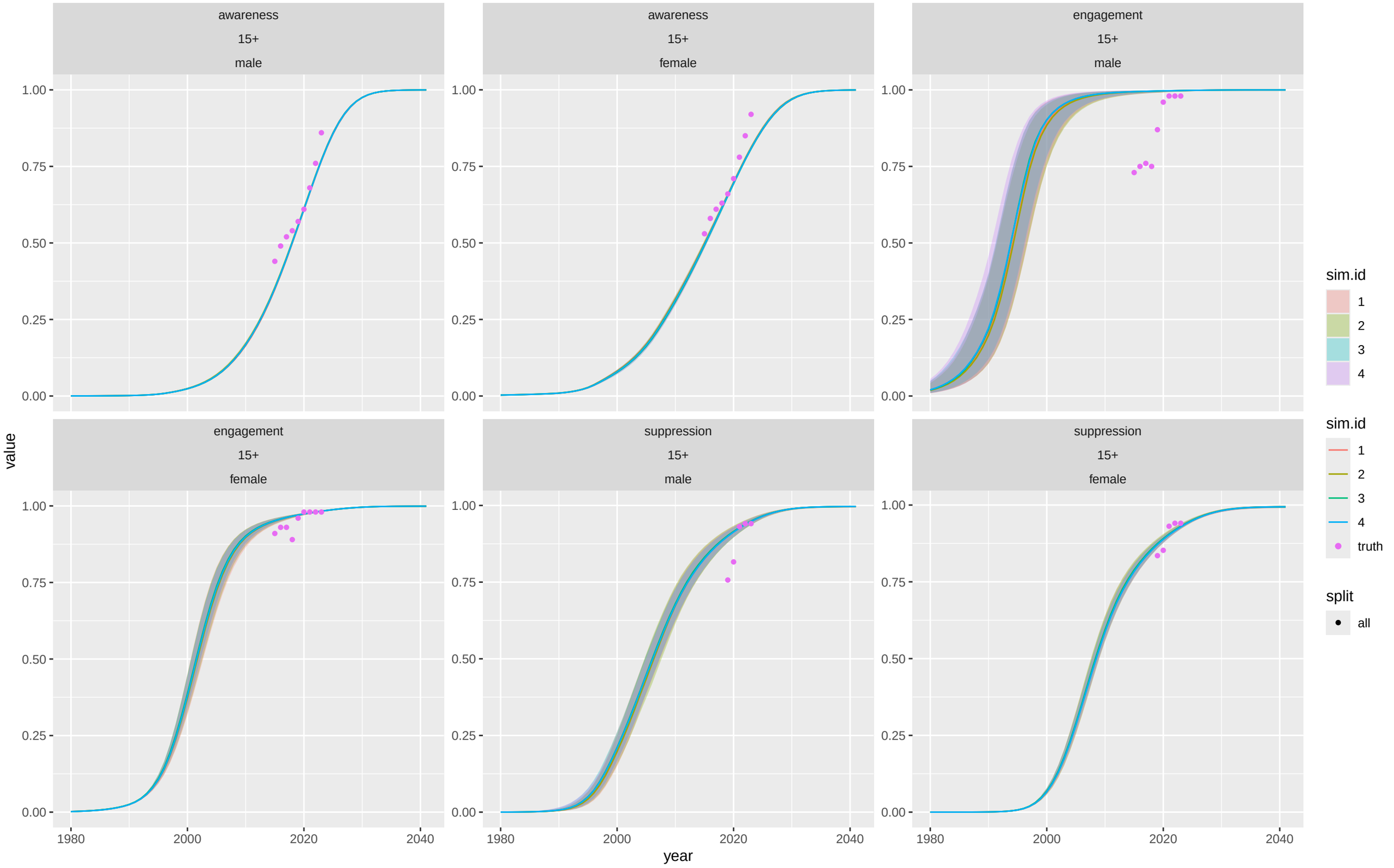
Nigeria step 2: population.by.age (one color per chain)



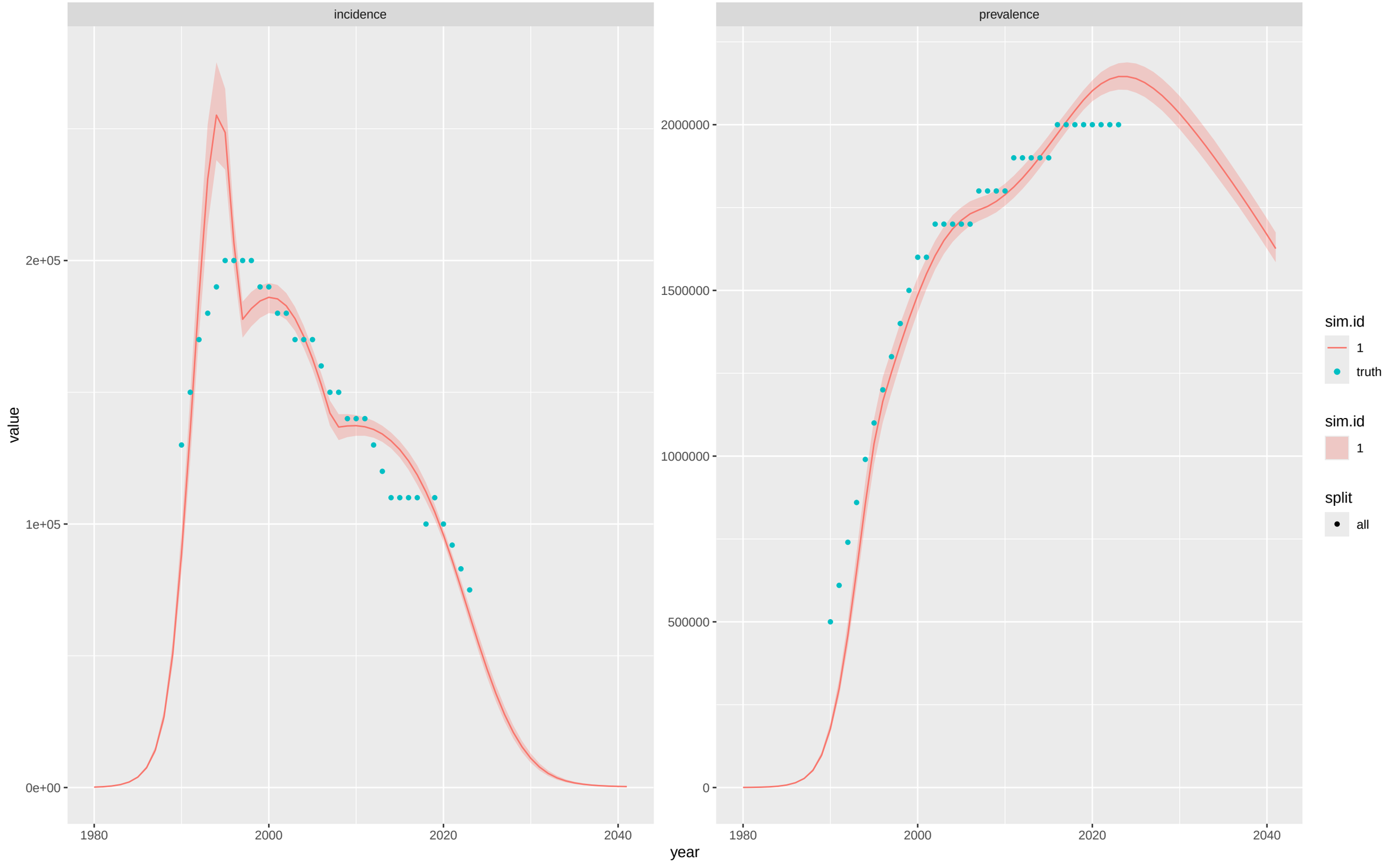
Nigeria step 2: cascade (one color per chain)



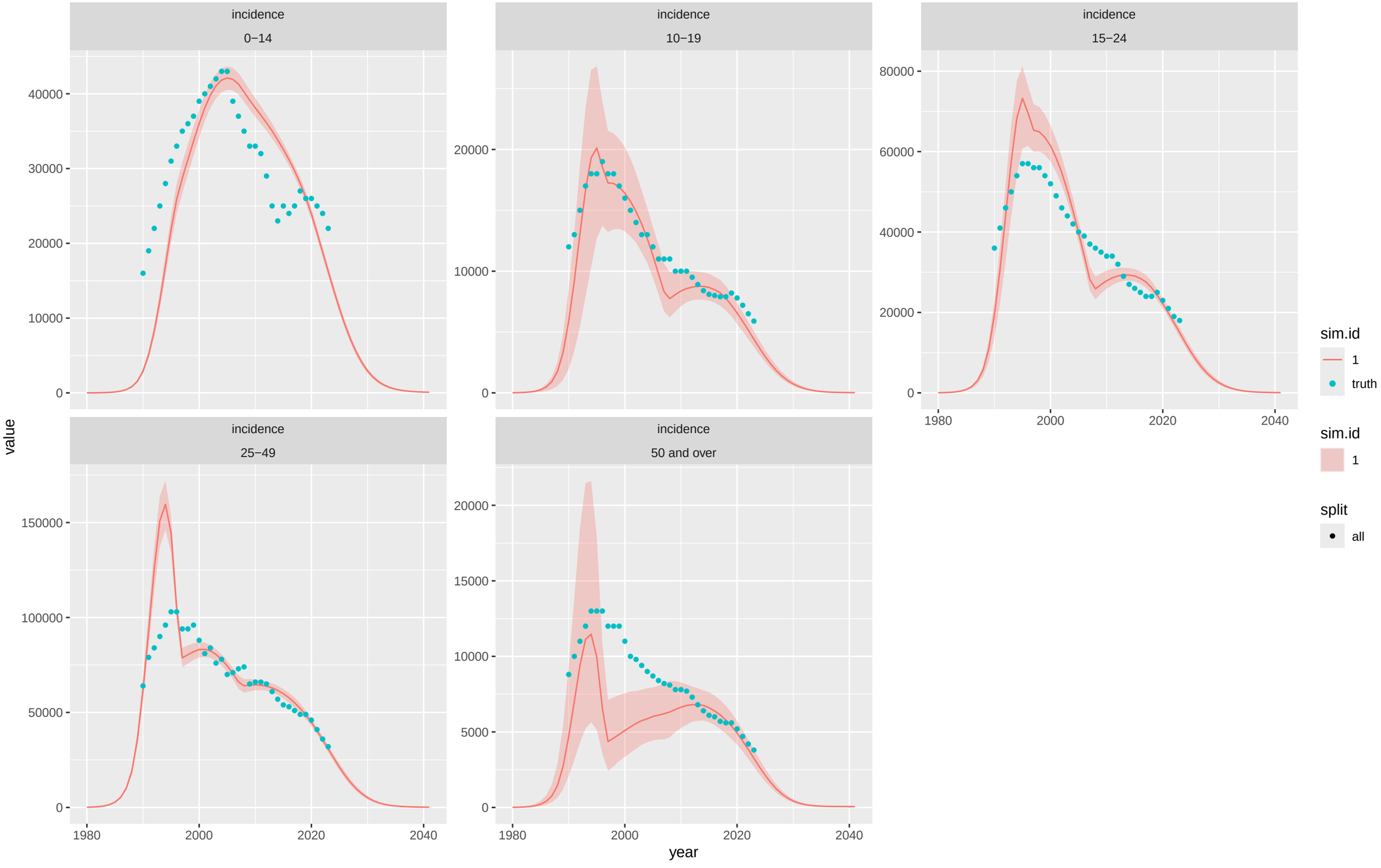
Nigeria step 2: cascade.by.age.sex (one color per chain)



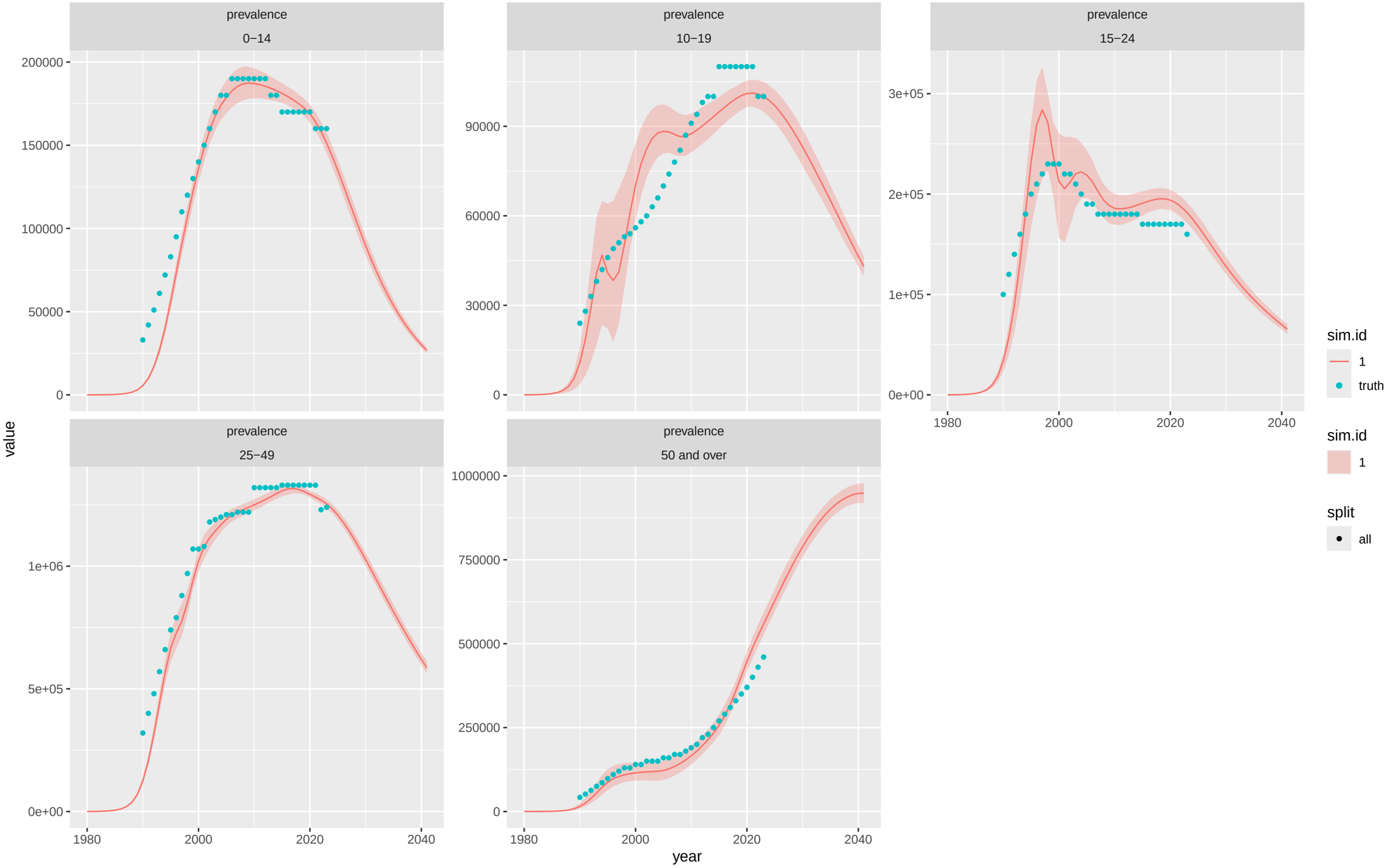
Nigeria: fit (merged Step 2 simset)



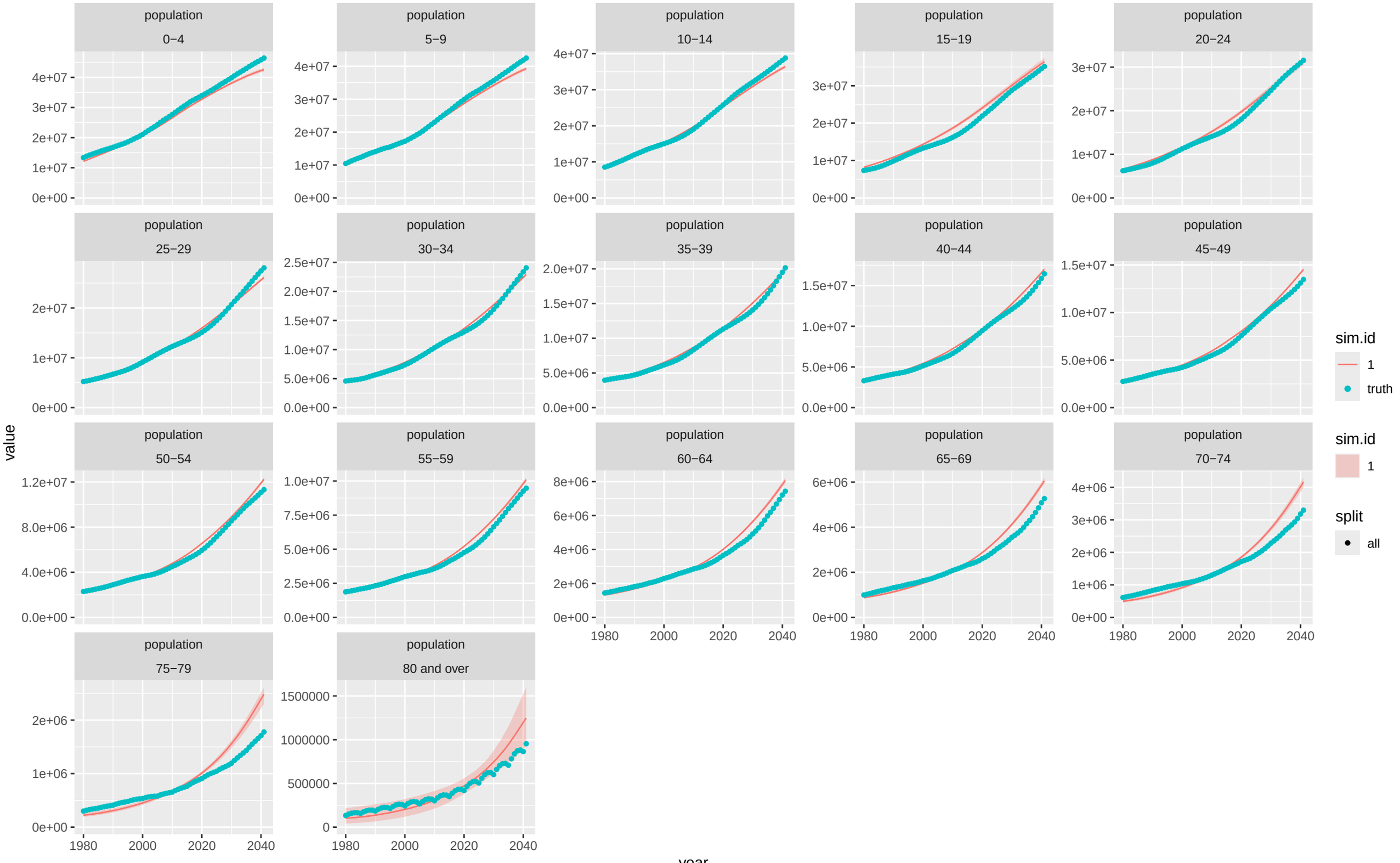
Nigeria: incidence.by.age (merged Step 2 simset)



Nigeria: prevalence.by.age (merged Step 2 simset)



Nigeria: population.by.age (merged Step 2 simset)



Nigeria: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

sim.id

1

split

all

Nigeria: cascade.by.age.sex (merged Step 2 simset)

